Buck-Boost DC–DC Converter

Steady-State Analysis: Inductor Current,
Output Current, and Duty-Cycle Relationships

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April 17, 2026

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1 Introduction

The *buck-boost* converter is a non-isolated, single-switch DC–DC topology that can produce an output voltage either lower or higher than its input, with a polarity inversion. It is widely used in battery management systems, LED drivers, and portable power supplies where the source voltage may swing above and below the regulated output.

This document develops the steady-state relationships between:

- the duty cycle D,
- the inductor current waveform $i_L(t)$, and
- the DC output current I_{out}.

Both Continuous Conduction Mode (CCM) and Discontinuous Conduction Mode (DCM) are treated, together with the boundary condition that separates them.

2 Circuit Topology and Operating Principle

2.1 Schematic

The canonical non-inverting representation places the load on the negative rail; in practice the output voltage V_{out} is negative with respect to the common ground when the switch is referenced to ground. Figure 1 shows the standard topology.

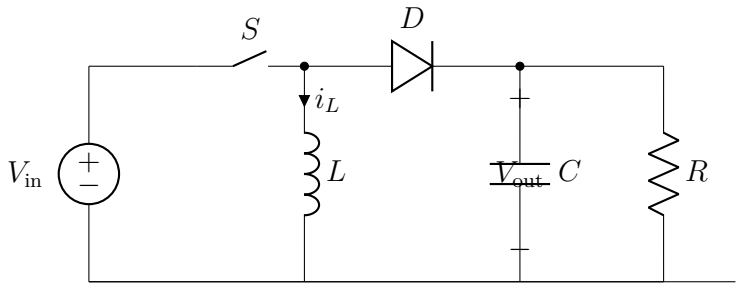


Figure 1: Buck-boost converter schematic. During the ON interval, S closes and L stores energy from V_{in}; the diode D is reverse-biased. During the OFF interval, S opens and L drives D into conduction, transferring energy to the output. The output voltage V_{out} is negative with respect to the input common ground.

2.2 Two Operating Intervals

The switch S operates at frequency $f_s = 1/T_s$ with duty cycle $D \in (0, 1)$.

Interval 1 — Switch ON ($0 \le t < DT_s$). The switch closes; the diode is reverse-biased. The source V_{in} is applied directly across the inductor:

$$v_L = V_{\text{in}}, \qquad \frac{di_L}{dt} = \frac{V_{\text{in}}}{L} > 0. \tag{1}$$

Energy is stored in L; the output capacitor C supplies the load alone.

Interval 2 — Switch OFF ($DT_s \le t < T_s$). The switch opens; the inductor forces the diode into conduction. The inductor now sees $-V_{\text{out}}$ (the capacitor voltage, which is negative):

$$v_L = V_{\text{out}} \ (< 0), \qquad \frac{di_L}{dt} = \frac{V_{\text{out}}}{L} < 0. \tag{2}$$

Energy is transferred from L to the output.

3 Voltage Conversion Ratio (CCM)

Applying the **volt-second balance** on the inductor over one switching period (net flux change must be zero in steady state):

$$V_{\text{in}} \cdot DT_s + V_{\text{out}} \cdot (1-D)T_s = 0. \tag{3}$$

Solving for the conversion ratio $M(D)$:

$$\boxed{M(D) = \frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{D}{1-D}.} \tag{4}$$

The negative sign confirms the polarity inversion. For $D = 0.5$, $|V_{\text{out}}| = V_{\text{in}}$; for $D > 0.5$, the converter boosts; for $D < 0.5$, it bucks.

4 Inductor Current Waveform and Ripple

4.1 Peak-to-Peak Ripple

From (1), the inductor current rises linearly during the ON interval:

$$\Delta I_L = \frac{V_{\text{in}}}{L}\, DT_s = \frac{V_{\text{in}}\, D}{L\, f_s}. \tag{5}$$

This can equally be expressed using the OFF interval and (2):

$$\Delta I_L = \frac{|V_{\text{out}}|}{L}(1-D)T_s, \tag{6}$$

which is consistent with (5) through (4).

4.2 Average Inductor Current Versus Output Current

This is the *central relationship* of this document.

In CCM, the inductor current i_L flows *only* through the diode during the OFF interval, because the switch is open. The output capacitor has zero average current in steady state; therefore *all* DC current to the load comes from the inductor through the diode during $(1-D)T_s$:

$$I_{\text{out}} = \frac{1}{T_s}\int_{DT_s}^{T_s} i_L(t)\, dt = (1-D)\,\langle I_L \rangle. \tag{7}$$

Solving for the average inductor current:

$$\boxed{\langle I_L \rangle = \frac{I_{\text{out}}}{1-D}.} \tag{8}$$

Key insight: Because $D \in (0,1)$, we always have $\langle I_L \rangle > I_{\text{out}}$. As $D \to 1$ the inductor must carry arbitrarily large average current to deliver the same I_{out}, which is a fundamental limitation of the buck-boost topology at high duty cycles.

4.3 Peak and Valley Values

$$I_{L,\max} = \langle I_L \rangle + \frac{\Delta I_L}{2} = \frac{I_{\text{out}}}{1-D} + \frac{V_{\text{in}}\, D}{2L\, f_s}, \tag{9}$$

$$I_{L,\min} = \langle I_L \rangle - \frac{\Delta I_L}{2} = \frac{I_{\text{out}}}{1-D} - \frac{V_{\text{in}}\, D}{2L\, f_s}. \tag{10}$$

The converter operates in CCM as long as $I_{L,\min} \geq 0$, i.e. $\langle I_L \rangle \geq \Delta I_L/2$.

5 Boundary Between CCM and DCM

The *critical* (boundary) inductance L_B is defined by $I_{L,\min} = 0$, which gives $\langle I_L \rangle = \Delta I_L/2$:

$$\frac{I_{\text{out}}}{1-D} = \frac{V_{\text{in}}\, D}{2L_B\, f_s}. \tag{11}$$

Substituting $V_{\text{in}} = |V_{\text{out}}|(1-D)/D$ from (4):

$$\boxed{L_B = \frac{(1-D)^2 R}{2D\, f_s},} \tag{12}$$

where $R = |V_{\text{out}}|/I_{\text{out}}$ is the load resistance. If $L > L_B$ the converter is in CCM; if $L < L_B$ it enters DCM.

Figure 2 illustrates the inductor current waveform at the CCM/DCM boundary and in full CCM.

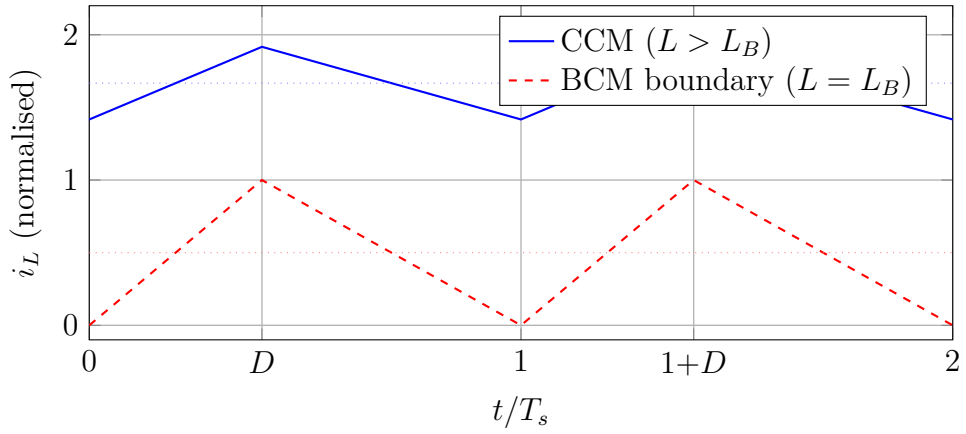


Figure 2: Inductor current waveforms for $D = 0.4$. The blue trace shows continuous conduction mode (CCM); the red dashed trace shows the boundary conduction mode (BCM). Dotted lines indicate the respective average values $\langle I_L \rangle$.

6 Discontinuous Conduction Mode (DCM)

In DCM the inductor current reaches zero before the end of the period and remains there for a third interval $\delta_3 T_s$, with $D + \delta_2 + \delta_3 = 1$.

6.1 Voltage Conversion Ratio in DCM

Charge balance on the output capacitor requires:

$$I_{\text{out}} = \frac{1}{2} I_{L,\max}\, \delta_2, \tag{13}$$

where δ_2 is the normalised duration of the diode conduction interval. Volt-second balance on L gives $\delta_2 = D V_{\text{in}}/|V_{\text{out}}|$, and $I_{L,\max} = V_{\text{in}} D T_s / L$. Combining:

$$M_{\text{DCM}}(D, K) = \frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{1 + \sqrt{1 + 4K/D^2}}{2}, \qquad K = \frac{2L\, f_s}{R}. \tag{14}$$

For $K \to 0$ (light load, small L), $|M_{\text{DCM}}| \to \infty$, confirming that the output voltage tends to rise without bound in open loop — the well-known DCM instability of the buck-boost.

6.2 Average Inductor Current in DCM

The average inductor current over the full period is:

$$\langle I_L \rangle = \frac{1}{2} I_{L,\max}(D + \delta_2) = \frac{V_{\text{in}} D T_s}{2L}(D + \delta_2). \tag{15}$$

Only the δ_2 portion flows to the output, reproducing (13).

7 Summary of Key Relationships

Table 1 collects the most important steady-state expressions in CCM.

Table 1: Steady-state relationships of the buck-boost converter in CCM.

Quantity	**Expression**	**Equation**
Voltage gain	$V_{\text{out}}/V_{\text{in}} = -D/(1-D)$	(4)
Inductor ripple	$\Delta I_L = V_{\text{in}} D/(L f_s)$	(5)
Average I_L	$\langle I_L \rangle = I_{\text{out}}/(1-D)$	(8)
Peak I_L	$I_{L,\max} = I_{\text{out}}/(1-D) + \Delta I_L/2$	(9)
Valley I_L	$I_{L,\min} = I_{\text{out}}/(1-D) - \Delta I_L/2$	(10)
Critical inductance	$L_B = (1-D)^2 R/(2D f_s)$	(12)

Physical interpretation of $\langle I_L \rangle = I_{\text{out}}/(1-D)$

The result (8) has a transparent physical origin. Because the inductor is connected to the output *only* during the OFF interval $(1-D)T_s$, the DC output current I_{out} is the time-average of a waveform that equals i_L for fraction $(1-D)$ of each period and zero otherwise. Therefore the DC component of i_L must be scaled up by $1/(1-D)$ to account for the duty-cycle-dependent *availability* of the inductor.

This is in contrast with:

- the **buck converter**, where $\langle I_L \rangle = I_{\text{out}}$ (inductor in series with the load at all times), and
- the **boost converter**, where $\langle I_L \rangle = I_{\text{out}}/(1-D)$ with an identical formula but a different physical path — in the boost, the *switch* current, not the inductor current, is pulsed; the inductor itself is always in series with the input.

8 Design Example

Specification: $V_{\text{in}} = 12\,\text{V}$, $V_{\text{out}} = -15\,\text{V}$, $I_{\text{out}} = 1\,\text{A}$, $f_s = 100\,\text{kHz}$, maximum ripple $\Delta I_L/\langle I_L \rangle \le 20\%$.

Duty cycle. From (4):

$$\frac{15}{12} = \frac{D}{1-D} \;\Rightarrow\; D = \frac{15}{15+12} = \frac{15}{27} \approx 0.556.$$

Average inductor current. From (8):

$$\langle I_L \rangle = \frac{1}{1-0.556} = 2.25\,\text{A}.$$

Inductance for target ripple. Allowed ripple: $\Delta I_L \leq 0.20 \times 2.25 = 0.45\,\text{A}$. From (5):

$$L \geq \frac{V_{\text{in}}\, D}{\Delta I_L\, f_s} = \frac{12 \times 0.556}{0.45 \times 10^5} \approx 148\,\mu\text{H}.$$

Choose $L = 150\,\mu\text{H}$.

Peak inductor current. From (9):

$$I_{L,\max} = 2.25 + \frac{0.45}{2} = 2.475\,\text{A}.$$

The inductor core must be rated for at least 2.48 A without saturation.

Critical inductance check. From (12), with $R = 15/1 = 15\,\Omega$:

$$L_B = \frac{(1 - 0.556)^2 \times 15}{2 \times 0.556 \times 10^5} \approx 26.6\,\mu\text{H}.$$

Since $L = 150\,\mu\text{H} \gg L_B = 26.6\,\mu\text{H}$, the converter is safely in CCM at full load.

9 Conclusions

The buck-boost converter inverts polarity and achieves voltage gain $|M| = D/(1-D)$ in CCM. The central design equation linking inductor current to output current is:

$$\langle I_L \rangle = \frac{I_{\text{out}}}{1 - D},$$

which arises because the inductor delivers charge to the output only during the OFF fraction $(1-D)$ of each period. Consequently, the average inductor current always exceeds the output current, and becomes increasingly large as $D \to 1$, imposing a practical upper limit on the achievable step-up ratio in a lossless design. In DCM the output voltage becomes load-dependent and tends to rise unboundedly at light load, underscoring the need for closed-loop voltage regulation in practical implementations.